Laboratory Evaluation Report

Developing an AI-Powered Image Classification Web Application Using JavaScript

Course	ICS106 - Application Development and Emerging Technologies
Activity	Laboratory 10 - Smart Object Recognition Application
Student	Lelius Lawas
Date	11 November 2026
Models	Model V1 (baseline) and Model V2 (improved), Google Teachable Machine
Model V1 URL	https://teachablemachine.withgoogle.com/models/-srVGQve0/
Model V2 URL	https://teachablemachine.withgoogle.com/models/WRjiyFqYc/
Headline result	Overall accuracy rose from 30 / 36 (83.3%) with V1 to 33 / 36 (91.7%) with V2 on the same 36 held-out test images.

1. Application Overview

The **Smart Object Recognition Application** is a single-page web app that loads two Teachable Machine image classification models, runs them on uploaded images, and records how reliably each one recognises everyday waste objects when the object is rotated, dimly lit, in front of a busy background, or far from the camera. Its purpose is not only to demonstrate a working classifier, but to expose weaknesses, support an improvement (Model V2), and prove the effect of that improvement with identical test cases.

Object classes. The classifier recognises three categories: **PAPER**, **PLASTIC BOTTLE**, and **ALUMINUM CAN**.

Features (mapped to the functional requirements).

ID	Requirement	How the application meets it
FR-01	Load a valid model	Loads a published Teachable Machine URL (or a local model folder) with model.json and metadata.json
FR-02	Upload an image	Drag-and-drop or file picker with image preview
FR-03	Display predicted category	The top class is shown in large text
FR-04	Confidence for all classes	A bar and percentage for every class
FR-05	Highest-scoring class	Predictions are sorted; the first is marked (highest)
FR-06	Show model version	Header tag and "Active model" field show V1 or V2
FR-07	Replace the model	V1 and V2 load independently; the active model can be switched
FR-08	Handle invalid input	Unsupported or oversized files and bad or unreachable model URLs show an error and keep the working model
FR-09	Webcam recognition	Live classification from the camera using the active model
FR-10	Uncertainty warning	Warns when the top score is below a set threshold (default 70%) or the top two classes are within 15 points

Testing tools added to support the lab. A condition generator makes the six required versions of a photo, both models classify every version, results are logged side by side with an Improved / Regressed / Same column, and per-condition accuracy and CSV exports are produced automatically.

2. Application Architecture

File	Role
index.html	Page structure: model loaders (V1/V2), image upload, results panel, condition check, test log, webcam
style.css	Layout and visual design (light theme; green = correct/loaded, red = wrong/failed, orange = uncertain)
script.js	Model loading, classification, uncertainty logic, condition generation, logging, accuracy summary, CSV export, webcam

Model interaction. The page loads TensorFlow.js 1.3.1 and the Teachable Machine image library 0.8.3 from a CDN. For each model, `tmImage.load(url + "model.json", url + "metadata.json")` creates a model object, and `model.predict(element)` returns a probability for every class. The application sorts the results by probability, takes the highest as the predicted class, and renders all scores. A new model replaces its slot only after it has loaded successfully, so a failed load never breaks a working model.

Fair comparison. Both models stay loaded together. Each classification or condition test runs V1 and V2 on exactly the same image data, and the generated conditions are deterministic (seeded background pattern, fixed sizes), so differences between V1 and V2 come from the models, not the inputs.

Data flow: upload image → preview loads → (optional) six condition versions drawn on canvases → both models predict → results rendered → recorded to the log (stored in the browser) → summary and CSV export.

3. Baseline Model Evaluation (Model V1)

Test set. 36 held-out images = 6 conditions x 3 classes x 2 images. They come from six source photographs (two per class, named 01-paper, 02-paper, 01-plastic-bottle, 02-plastic-bottle, 01-aluminum-can, 02-aluminum-can). Each photo was turned into the six required conditions. A result counts as correct only when the highest-scoring class equals the known actual class. No results were discarded.

Result: Model V1 classified 30 of 36 images correctly (83.3%).

Table 1. Baseline Model V1 Testing Log

Test ID	Condition	Actual Class	Predicted Class	Confidence	Result
T01	Upright	PAPER	PAPER	100%	Correct
T02	Rotated 90°	PAPER	PAPER	100%	Correct
T03	Rotated 180°	PAPER	PAPER	100%	Correct
T04	Low lighting	PAPER	PAPER	100%	Correct
T05	Complex background	PAPER	PAPER	98.9%	Correct
T06	Increased distance	PAPER	PAPER	99.5%	Correct
T07	Upright	PAPER	PAPER	93.9%	Correct
T08	Rotated 90°	PAPER	PAPER	55.8%	Correct
T09	Rotated 180°	PAPER	PAPER	52.5%	Correct
T10	Low lighting	PAPER	PAPER	99.6%	Correct
T11	Complex background	PAPER	PAPER	82.2%	Correct
T12	Increased distance	PAPER	PAPER	84.4%	Correct
T13	Upright	PLASTIC BOTTLE	PLASTIC BOTTLE	100%	Correct
T14	Rotated 90°	PLASTIC BOTTLE	PLASTIC BOTTLE	95.2%	Correct
T15	Rotated 180°	PLASTIC BOTTLE	PLASTIC BOTTLE	97.7%	Correct
T16	Low lighting	PLASTIC BOTTLE	PLASTIC BOTTLE	99.4%	Correct
T17	Complex background	PLASTIC BOTTLE	PLASTIC BOTTLE	98.1%	Correct
T18	Increased distance	PLASTIC BOTTLE	PAPER	77.1%	Wrong
T19	Upright	PLASTIC BOTTLE	PLASTIC BOTTLE	55.4%	Correct
T20	Rotated 90°	PLASTIC BOTTLE	PAPER	62.4%	Wrong
T21	Rotated 180°	PLASTIC BOTTLE	PLASTIC BOTTLE	93.8%	Correct
T22	Low lighting	PLASTIC BOTTLE	PAPER	94%	Wrong
T23	Complex background	PLASTIC BOTTLE	PAPER	86.2%	Wrong
T24	Increased distance	PLASTIC BOTTLE	ALUMINUM CAN	78.9%	Wrong
T25	Upright	ALUMINUM CAN	ALUMINUM CAN	99.9%	Correct
T26	Rotated 90°	ALUMINUM CAN	ALUMINUM CAN	99.7%	Correct
T27	Rotated 180°	ALUMINUM CAN	ALUMINUM CAN	99.9%	Correct
T28	Low lighting	ALUMINUM CAN	ALUMINUM CAN	99.2%	Correct
T29	Complex background	ALUMINUM CAN	ALUMINUM CAN	96.4%	Correct
T30	Increased distance	ALUMINUM CAN	ALUMINUM CAN	90%	Correct
T31	Upright	ALUMINUM CAN	ALUMINUM CAN	99.9%	Correct
T32	Rotated 90°	ALUMINUM CAN	ALUMINUM CAN	99%	Correct
T33	Rotated 180°	ALUMINUM CAN	ALUMINUM CAN	97.2%	Correct
T34	Low lighting	ALUMINUM CAN	ALUMINUM CAN	98.6%	Correct
T35	Complex background	ALUMINUM CAN	ALUMINUM CAN	93.6%	Correct
T36	Increased distance	ALUMINUM CAN	PAPER	92.6%	Wrong

Table 1a. V1 accuracy per condition

Condition	V1 correct / 6	V1 accuracy
Upright	6 / 6	100.0%
Rotated 90°	5 / 6	83.3%
Rotated 180°	6 / 6	100.0%
Low lighting	5 / 6	83.3%
Complex background	5 / 6	83.3%
Increased distance	3 / 6	50.0%
Overall	**30 / 36**	**83.3%**

Table 1b. V1 accuracy per class

Class	V1 correct / 12	V1 accuracy
PAPER	12 / 12	100.0%
PLASTIC BOTTLE	7 / 12	58.3%
ALUMINUM CAN	11 / 12	91.7%

4. Identified Problems

Model V1 was reliable on upright objects (6 / 6) and on 180° rotation (6 / 6), but its accuracy dropped when the object was shrunk to imitate distance. The weaknesses were also concentrated in one class.

Table 2a. Diagnostic record (Task 5)

Component	Student Response
Lowest-performing condition	Increased distance: 3 / 6 correct (50.0%).
Second-lowest-performing condition	A three-way tie at 5 / 6 (83.3%): Rotated 90°, Low lighting, and Complex background. In each case the errors came from the same plastic-bottle photo (02-plastic-bottle), so they are treated together as "plastic bottle under changed conditions".
Most frequently confused classes	PLASTIC BOTTLE predicted as PAPER (4 of V1's 6 errors). The other errors were PLASTIC BOTTLE as ALUMINUM CAN (1) and ALUMINUM CAN as PAPER (1).
Possible causes	(1) PLASTIC BOTTLE is the weakest class: 7 / 12 versus 12 / 12 for PAPER and 11 / 12 for ALUMINUM CAN (Table 1b). The 02-plastic-bottle photo was only 2 / 6 correct, so the model likely lacks varied bottle examples (clear, tilted, dim, cluttered). (2) At reduced size the object fills only about a third of the frame, so the model sees few object pixels and relies on background and colour. (3) Some correct predictions were already shaky (e.g. 52.5% and 55.8% on 02-paper at 180° and 90°, 55.4% on 02-plastic-bottle upright), showing little margin.
Proposed model modification	Add varied training samples for every class, with extra emphasis on PLASTIC BOTTLE and on small / distant objects, tilted or sideways objects, dim lighting and cluttered backgrounds, keeping the three class definitions and the same training settings.
Expected improvement	Higher accuracy on Increased distance and the plastic-bottle failures, with no loss on upright images.

V1 errors in detail (all six are model classification errors, not application errors):

Test	Condition	Actual	V1 predicted	Conf.
T18	Increased distance	PLASTIC BOTTLE	PAPER	77.1%
T20	Rotated 90°	PLASTIC BOTTLE	PAPER	62.4%
T22	Low lighting	PLASTIC BOTTLE	PAPER	94%
T23	Complex background	PLASTIC BOTTLE	PAPER	86.2%
T24	Increased distance	PLASTIC BOTTLE	ALUMINUM CAN	78.9%
T36	Increased distance	ALUMINUM CAN	PAPER	92.6%

Note that these wrong answers were given with 62% to 94% confidence, so the uncertainty warning alone would not have caught most of them (see guide question 9).

5. Model Improvement

Improvement hypothesis. Adding varied training images of each object - especially plastic bottles - in different orientations, at greater distances, under dim light and in front of cluttered backgrounds will raise accuracy on the Increased distance condition and on the plastic-bottle failures without reducing accuracy on upright images.

Retraining procedure. Model V1 was preserved (its URL is recorded below). Teachable Machine was reopened, the existing dataset was reviewed, additional samples were collected for the weak conditions, the model was retrained with the original three classes, and it was published as Model V2. The additional samples were not copies or modifications of the 36 reserved test images.

Table 3. Before-and-after training record

Class	Samples added for V2	Weaknesses addressed
PAPER	At least 15	Rotated and distant paper; cluttered backgrounds
PLASTIC BOTTLE	At least 15	Weakest class in V1 (7 / 12): distant, sideways, dim and cluttered bottle photos
ALUMINUM CAN	At least 15	Distant cans (T36 error in V1)
Model V1 URL	https://teachablemachine.withgoogle.com/models/-srVGQve0/	
Model V2 URL	https://teachablemachine.withgoogle.com/models/WRjiyFqYc/	

6. Comparative Evaluation (V1 vs V2)

Both models were run on the exact same 36 images. **Model V2 classified 33 of 36 correctly (91.7%), compared with 30 of 36 (83.3%) for V1: a gain of 3 images (+8.3 percentage points).** Four predictions improved, one regressed, and 31 were unchanged (29 correct in both, 2 wrong in both).

Table 4. Model V1 and V2 comparative testing

Condition	V1 correct / 6	V2 correct / 6	Change
Upright	6 / 6	6 / 6	0
Rotated 90°	5 / 6	6 / 6	+1
Rotated 180°	6 / 6	6 / 6	0
Low lighting	5 / 6	6 / 6	+1
Complex background	5 / 6	5 / 6	0
Increased distance	3 / 6	4 / 6	+1
Overall / 36	**30 / 36**	**33 / 36**	**+3**

Correct predictions out of 6 per condition (grey = V1, blue = V2)

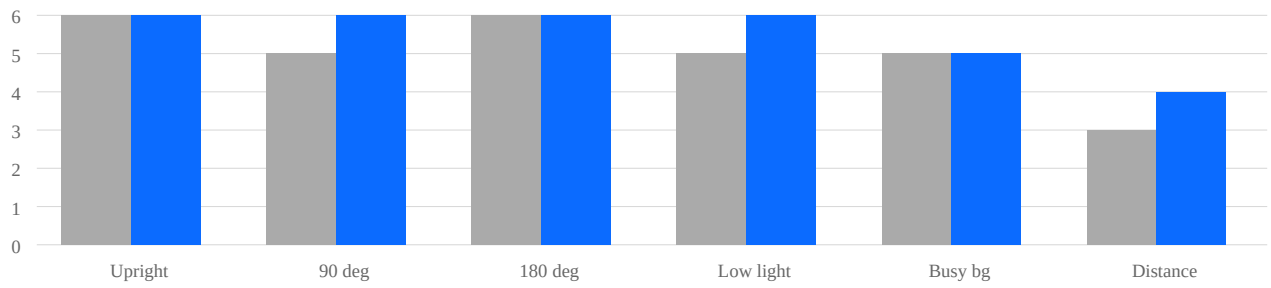

Table 4a. Accuracy per class

Class	V1 correct / 12	V2 correct / 12	Change
PAPER	12 / 12	11 / 12	-1
PLASTIC BOTTLE	7 / 12	10 / 12	+3
ALUMINUM CAN	11 / 12	12 / 12	+1

Improvements, regressions and unchanged behaviour

Test	Image	V1	V2	Change
T12	02-paper [far]	PAPER 84.4%	ALUMINUM CAN 54.3%	Regressed
T18	01-plastic-bottle [far]	PAPER 77.1%	PLASTIC BOTTLE 52.5%	Improved
T20	02-plastic-bottle [rot90]	PAPER 62.4%	PLASTIC BOTTLE 79.7%	Improved
T22	02-plastic-bottle [dark]	PAPER 94%	PLASTIC BOTTLE 63.7%	Improved
T23	02-plastic-bottle [busy]	PAPER 86.2%	PAPER 84.7%	Same (wrong)
T24	02-plastic-bottle [far]	ALUMINUM CAN 78.9%	ALUMINUM CAN 93%	Same (wrong)
T36	02-aluminum-can [far]	PAPER 92.6%	ALUMINUM CAN 60.2%	Improved

- **Improved (4):** T18 plastic bottle at distance, T20 plastic bottle rotated 90°, T22 plastic bottle in low light, T36 aluminum can at distance. All four were V1 errors; three involve the plastic bottle.
- **Regressed (1):** T12, paper at a distance, was correct in V1 (84.4%) but V2 called it ALUMINUM CAN (54.3%). This is the side effect of the retraining and it is reported, not hidden.
- **Still wrong in both (2):** T23 plastic bottle with a complex background (predicted PAPER) and T24 plastic bottle at a distance (predicted ALUMINUM CAN, with 93.0% confidence in V2).
- **Unchanged and correct (29):** including every upright image (6 / 6 in both models). The improvements for non-upright images did not reduce upright accuracy.

Several of the gains are low-confidence: V2 got T18 right at 52.5%, T36 at 60.2% and T22 at 63.7%. These are correct answers with little margin, and with a 70% threshold the uncertainty warning would flag them.

7. Functional Testing

The application was tested for its functional behaviour in addition to the 36 classification cases. Each test was run in the app and its result recorded as Pass or Fail. Failures of the model to classify an image correctly are not application failures and are covered in Section 6.

Test ID	Functional Test	Expected Behavior	Result	Correction
FT-01	Open the application	Interface loads correctly	**Pass**	None required
FT-02	Initialize Model V1	Model loads successfully	**Pass**	None required
FT-03	Upload a supported image	Image preview appears	**Pass**	None required
FT-04	Classify an image	Predicted class is displayed	**Pass**	None required
FT-05	Display confidence scores	All trained classes are shown	**Pass**	None required
FT-06	Load Model V2	Revised model loads successfully	**Pass**	None required
FT-07	Classify using Model V2	New model produces predictions	**Pass**	None required
FT-08	Upload an unsupported file	Application provides an error	**Pass**	None required
FT-09	Use an invalid model URL	Loading error is displayed	**Pass**	None required
FT-10	Classify before model loading	Application prevents or safely handles the operation	**Pass**	None required

Application errors versus model errors. The application handles invalid files, oversized files, bad model URLs, and classification before a model is loaded by showing a message instead of failing silently (FT-08 to FT-10). In the 36 image tests the application behaved correctly every time; every wrong answer was a prediction made by the model.

Source of the improvement. The accuracy gain came from changing the training data (Model V2). The JavaScript was not changed between the V1 and V2 test runs: the same code and the same test images were used for both.

8. Conclusion

Improvements. Retraining raised overall accuracy from 83.3% to 91.7% (30 to 33 of 36). The largest gains were on the PLASTIC BOTTLE class (7 / 12 to 10 / 12) and on Rotated 90° (5 to 6), Low lighting (5 to 6), and Increased distance (3 to 4). Upright and 180° accuracy stayed perfect.

Remaining limitations.

- Increased distance is still the weakest condition (4 / 6) and Complex background did not improve (5 / 6).
- PAPER got slightly worse (12 / 12 to 11 / 12): the T12 regression shows retraining can trade accuracy in one place for another.
- The two images still wrong in V2 are both plastic bottle photos, one predicted at 93% confidence, so confidence alone cannot be trusted.
- The test set is small: 36 images from six source photos. The six conditions of one photo are not independent, so a change of three images should be treated as encouraging evidence, not proof.
- The lighting, background and distance conditions were simulated from one photo (darkened overlay, shrunk onto a pattern or plain background), so real-world conditions may behave differently.

Recommendations.

- Add more real photographs per class taken at different distances, in real dim light and in real cluttered scenes, and collect more plastic bottle varieties (clear, coloured, tilted).
- Test with many more independent photos per condition and report accuracy per class as well as overall.
- Use the uncertainty warning (or a rejection threshold) so low-confidence predictions are shown as "not sure" instead of a class name.
- Re-run the same 36 test images after every retraining to catch regressions such as T12.

Guide Questions

1. Why can an image classifier correctly identify an upright object but fail to identify the same object when rotated?

A Teachable Machine model learns patterns of colour, shape and position from the pixels in its training images, and most training images show the object upright. A rotated object puts the same features in different places and at different angles, so the learned patterns no longer line up. The model does not understand what the object is; it matches patterns, and rotation changes the pattern. In our results the V1 plastic bottle photo 02 failed when turned 90° (T20, predicted PAPER at 62.4%) even though it was classified correctly upright (T19, 55.4%).

2. Which orientation, 90° or 180°, produced the greatest classification difficulty? What evidence supports your answer?

For V1, 90° was slightly harder: 5 / 6 correct versus 6 / 6 at 180°; the one 90° error was T20. For V2, both were 6 / 6. The evidence is thin: the difference is one image. Both orientations also produced low-confidence correct answers on 02-paper in V1 (55.8% at 90° and 52.5% at 180°), so neither orientation was a strong weakness. Increased distance was clearly the harder condition (3 / 6 for V1).

3. How did lighting, distance, and background changes affect the classifier's performance?

V1: Low lighting 5 / 6, Complex background 5 / 6 and Increased distance 3 / 6. Distance hurt the most because shrinking the object leaves few object pixels (T18 and T36 became PAPER, T24 became ALUMINUM CAN). Lighting and background errors came from the 02-plastic-bottle photo (T22 and T23, both predicted PAPER). V2: Low lighting 6 / 6, Complex background 5 / 6, Increased distance 4 / 6. Lighting was fixed, distance improved by one image, background did not improve.

4. What changes were made to the training dataset, and how were these changes intended to address the classification problems?

Additional varied training samples were added to each class (at least 15 per class) to cover the weak cases found in V1: distant or small objects, rotated objects, dim lighting and cluttered backgrounds, with extra attention on PLASTIC BOTTLE, the weakest class (7 / 12). The class definitions and training settings were kept. The intent was to show the model more variety so it relies on the object itself rather than on position, lighting or background. The additions per class are summarised in Table 3.

5. Did Model V2 improve recognition across every test condition? If not, which conditions remained problematic?

No. V2 improved Rotated 90° (5 to 6), Low lighting (5 to 6) and Increased distance (3 to 4). Upright (6 / 6), Rotated 180° (6 / 6) and Complex background (5 / 6) did not change. Increased distance (4 / 6) and Complex background (5 / 6) remained problematic, and T23 and T24 were wrong in both models.

6. Why is it important to use the same held-out test images when comparing two versions of a model?

If the two models were tested on different images, any difference in score could come from easier or harder images rather than a better model. Using the same images changes only one thing, the model. Held-out images (not used in training) also show real generalisation instead of memorisation. The same 36 images let us identify exact improvements (T18, T20, T22, T36) and the regression (T12).

7. Can a model become better at recognizing rotated objects while becoming less accurate in other conditions? Explain using your results.

Yes. Training changes shift what the model has learned, which can help some cases and hurt others. V2 improved rotated and dark plastic bottle images, but the paper photo 02-paper at a distance went from correct (T12, 84.4%) in V1 to wrong (ALUMINUM CAN at 54.3%) in V2, and the PAPER class dropped from 12 / 12 to 11 / 12. Upright accuracy stayed at 6 / 6, so this particular retraining did not hurt upright recognition.

8. What is the difference between an application programming error and a machine learning prediction error?

An application programming error is a fault in the code: for example, a button that does nothing, a model that never loads, a crash on an invalid file, or results shown for the wrong image. The program behaves differently from what it was written to do. A machine learning prediction error is when the code runs correctly but the model gives the wrong answer, for example predicting PAPER for a plastic bottle. In this lab, all 6 V1 mistakes and all 3 V2 mistakes were prediction errors; the application displayed exactly what the model returned.

9. Why is a high confidence score not sufficient evidence that a prediction is correct?

Confidence is how strongly the model favours its top class among the classes it knows, not a measure of how often it is right. A model can be sure and wrong when an image is unlike its training data. Our results show this: V1 predicted PAPER for a plastic bottle at 94.0% (T22) and 86.2% (T23), and V2 predicted ALUMINUM CAN for a plastic bottle at 93.0% (T24). Across V1's six errors the average confidence was 81.9%, only about 11 points below the average for correct answers (92.7%).

10. What additional improvements would be required before the application could be considered suitable for real-world deployment?

A much larger and more varied training set of real photographs (many objects of each type, brands, colours, backgrounds, lighting and distances), a larger independent test set with accuracy reported per class and per condition, an "unknown / not sure" option, a rejection threshold for low-confidence results, protection against objects outside the three classes, checks on the webcam pipeline under real conditions, and ongoing monitoring and retraining after deployment.

Appendix A. Full V1 and V2 Results (same 36 test images)

Green = correct, red = wrong. BOTTLE = PLASTIC BOTTLE, CAN = ALUMINUM CAN. Source files: V1-test-results.xlsx and V2-test-results.xlsx.

Test	Condition	Image	Actual	V1	V2	Change
T01	Upright	01-paper [upright]	PAPER	PAPER 100%	PAPER 100%	Same
T02	Rotated 90°	01-paper [rot90]	PAPER	PAPER 100%	PAPER 100%	Same
T03	Rotated 180°	01-paper [rot180]	PAPER	PAPER 100%	PAPER 100%	Same
T04	Low lighting	01-paper [dark]	PAPER	PAPER 100%	PAPER 100%	Same
T05	Complex background	01-paper [busy]	PAPER	PAPER 98.9%	PAPER 96.8%	Same
T06	Increased distance	01-paper [far]	PAPER	PAPER 99.5%	PAPER 98.4%	Same
T07	Upright	02-paper [upright]	PAPER	PAPER 93.9%	PAPER 98%	Same
T08	Rotated 90°	02-paper [rot90]	PAPER	PAPER 55.8%	PAPER 93.6%	Same
T09	Rotated 180°	02-paper [rot180]	PAPER	PAPER 52.5%	PAPER 85.6%	Same
T10	Low lighting	02-paper [dark]	PAPER	PAPER 99.6%	PAPER 99.7%	Same
T11	Complex background	02-paper [busy]	PAPER	PAPER 82.2%	PAPER 94.2%	Same
T12	Increased distance	02-paper [far]	PAPER	PAPER 84.4%	CAN 54.3%	Regressed
T13	Upright	01-plastic-bottle [upright]	BOTTLE	BOTTLE 100%	BOTTLE 100%	Same
T14	Rotated 90°	01-plastic-bottle [rot90]	BOTTLE	BOTTLE 95.2%	BOTTLE 88.2%	Same
T15	Rotated 180°	01-plastic-bottle [rot180]	BOTTLE	BOTTLE 97.7%	BOTTLE 95.9%	Same
T16	Low lighting	01-plastic-bottle [dark]	BOTTLE	BOTTLE 99.4%	BOTTLE 99.4%	Same
T17	Complex background	01-plastic-bottle [busy]	BOTTLE	BOTTLE 98.1%	BOTTLE 94.2%	Same
T18	Increased distance	01-plastic-bottle [far]	BOTTLE	PAPER 77.1%	BOTTLE 52.5%	Improved
T19	Upright	02-plastic-bottle [upright]	BOTTLE	BOTTLE 55.4%	BOTTLE 84.6%	Same
T20	Rotated 90°	02-plastic-bottle [rot90]	BOTTLE	PAPER 62.4%	BOTTLE 79.7%	Improved
T21	Rotated 180°	02-plastic-bottle [rot180]	BOTTLE	BOTTLE 93.8%	BOTTLE 99.3%	Same
T22	Low lighting	02-plastic-bottle [dark]	BOTTLE	PAPER 94%	BOTTLE 63.7%	Improved
T23	Complex background	02-plastic-bottle [busy]	BOTTLE	PAPER 86.2%	PAPER 84.7%	Same (wrong)
T24	Increased distance	02-plastic-bottle [far]	BOTTLE	CAN 78.9%	CAN 93%	Same (wrong)
T25	Upright	01-aluminum-can [upright]	CAN	CAN 99.9%	CAN 100%	Same
T26	Rotated 90°	01-aluminum-can [rot90]	CAN	CAN 99.7%	CAN 99.9%	Same
T27	Rotated 180°	01-aluminum-can [rot180]	CAN	CAN 99.9%	CAN 100%	Same
T28	Low lighting	01-aluminum-can [dark]	CAN	CAN 99.2%	CAN 99.9%	Same
T29	Complex background	01-aluminum-can [busy]	CAN	CAN 96.4%	CAN 97.7%	Same
T30	Increased distance	01-aluminum-can [far]	CAN	CAN 90%	CAN 99.6%	Same
T31	Upright	02-aluminum-can [upright]	CAN	CAN 99.9%	CAN 100%	Same
T32	Rotated 90°	02-aluminum-can [rot90]	CAN	CAN 99%	CAN 98.3%	Same
T33	Rotated 180°	02-aluminum-can [rot180]	CAN	CAN 97.2%	CAN 99.1%	Same
T34	Low lighting	02-aluminum-can [dark]	CAN	CAN 98.6%	CAN 99.6%	Same
T35	Complex background	02-aluminum-can [busy]	CAN	CAN 93.6%	CAN 97.7%	Same
T36	Increased distance	02-aluminum-can [far]	CAN	PAPER 92.6%	CAN 60.2%	Improved